

Amazon Brazil

SQL Analysis

A 14-query portfolio case study in e-commerce data analytics — seller performance, revenue trends, payments, delivery, and customer satisfaction.

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Data Analytics Certification Program • WsCube Tech

Q1 Orders by Seller State

1

Which Brazilian states have the most active sellers by order volume?

SQL QUERY

```
SELECT
    s.seller_state,
    COUNT(DISTINCT oi.order_id) AS total_orders
FROM sellers s
JOIN order_items oi
    ON s.seller_id = oi.seller_id
GROUP BY s.seller_state
ORDER BY total_orders DESC;
```

EXPLANATION

- Joins sellers with order_items on seller_id
- Counts distinct orders for each seller state
- Groups and sorts states by total order volume

EXPECTED OUTPUT

seller_state	total_orders
SP	45,230
RJ	8,120
MG	6,540

BUSINESS INSIGHT

Reveals which states host Amazon's most active seller base, guiding regional warehouse and logistics investment.

Q2 Monthly Product Revenue

2

How does revenue and order volume trend per product each month?

SQL QUERY

SELECT

```
DATE_FORMAT(o.order_purchase_timestamp, '%Y-%m') AS order_month,  
p.product_id,  
p.product_category_name,  
COUNT(oi.order_id) AS total_orders,  
SUM(oi.price) AS revenue
```

FROM

```
products p
```

JOIN

```
order_items oi  
ON p.product_id = oi.product_id
```

JOIN

```
orders o  
ON oi.order_id = o.order_id
```

GROUP BY

```
DATE_FORMAT(o.order_purchase_timestamp, '%Y-%m'),  
p.product_id,  
p.product_category_name
```

ORDER BY

```
order_month,  
revenue DESC;
```

EXPLANATION

- Formats the purchase timestamp into a year-month period
- Joins products, order_items, and orders together
- Aggregates order count and revenue by month and product

EXPECTED OUTPUT

order_month	product_id	revenue
2018-01	8b1f...	R\$12,450
2018-02	3a2c...	R\$9,870

BUSINESS INSIGHT

Enables trend analysis of product performance over time to guide inventory and marketing timing.

Q3 Payment Type Performance

3

Which payment methods do customers prefer, and what's the average spend?

SQL QUERY

```
SELECT
    payment_type,
    COUNT(order_id) AS total_orders,
    ROUND(AVG(payment_value), 2) AS average_order_value
FROM order_payments
GROUP BY payment_type
ORDER BY total_orders DESC;
```

EXPLANATION

- Groups all order payments by payment_type
- Counts total orders placed with each method
- Calculates the average payment value, rounded to 2 decimals

EXPECTED OUTPUT

payment_type	total_orders	avg_value
credit_card	76,795	163.32
boleto	19,784	145.03

BUSINESS INSIGHT

Highlights the dominant payment channel, informing checkout UX priorities and payment-partner negotiations.

Q4 Top Spending Customer

4

Who is Amazon Brazil's single highest-value customer?

SQL QUERY

```
SELECT
    c.customer_unique_id,
    SUM(op.payment_value) AS total_spent
FROM customers c
JOIN orders o
    ON c.customer_id = o.customer_id
JOIN order_payments op
    ON o.order_id = op.order_id
GROUP BY c.customer_unique_id
ORDER BY total_spent DESC
LIMIT 1;
```

EXPLANATION

- Joins customers, orders, and order_payments
- Sums total payment value per unique customer
- Sorts descending and returns only the top spender

EXPECTED OUTPUT

customer_unique_id	total_spent
8d50f...	R\$13,664.08

BUSINESS INSIGHT

Identifies VIP customers who qualify for loyalty rewards or personalized retention campaigns.

Q5 Review Score by Category

5

Which product categories earn the highest customer satisfaction?

SQL QUERY

```
SELECT
    p.product_category_name,
    ROUND(AVG(orv.review_score), 2) AS avg_review_score
FROM products p
JOIN order_items oi
    ON p.product_id = oi.product_id
JOIN order_reviews orv
    ON oi.order_id = orv.order_id
GROUP BY p.product_category_name
ORDER BY avg_review_score DESC;
```

EXPLANATION

- Joins products, order_items, and order_reviews
- Averages the review_score for each product category
- Sorts categories from highest to lowest rating

EXPECTED OUTPUT

product_category_name	avg_review_score
books	4.6
electronics	3.8

BUSINESS INSIGHT

Flags low-rated categories for quality review and highlights top categories worth promoting.

Q6 Orders by Customer & State

6

How many orders does each customer place, broken down by state?

SQL QUERY

```
SELECT
    c.customer_state,
    c.customer_unique_id,
    COUNT(o.order_id) AS total_orders
FROM customers c
JOIN orders o
    ON c.customer_id = o.customer_id
GROUP BY
    c.customer_state,
    c.customer_unique_id
ORDER BY
    c.customer_state,
    total_orders DESC;
```

EXPLANATION

- Joins customers with their placed orders
- Groups by customer state, then by individual customer
- Counts total orders per customer within each state

EXPECTED OUTPUT

customer_state	customer_id	total_orders
SP	7c1a...	4
SP	2b9f...	3

BUSINESS INSIGHT

Supports state-level customer segmentation for targeted regional marketing campaigns.

Q7 Sellers With Zero Orders

7

Which registered sellers have never fulfilled a single order?

SQL QUERY

```
SELECT
    s.seller_id,
    s.seller_city,
    s.seller_state
FROM sellers s
LEFT JOIN order_items oi
    ON s.seller_id = oi.seller_id
WHERE oi.order_id IS NULL;
```

EXPLANATION

- Left joins sellers to order_items
- Keeps only sellers with no matching order rows
- Returns seller ID, city, and state for follow-up

EXPECTED OUTPUT

seller_id	seller_city	seller_state
f8d9...	Curitiba	PR
a41c...	Recife	PE

BUSINESS INSIGHT

Flags inactive sellers for re-engagement, onboarding support, or account review.

Q8 Top 5 Revenue Categories

8

Which five product categories generate the most total revenue?

SQL QUERY

```
SELECT
    p.product_category_name,
    ROUND(SUM(oi.price), 2) AS total_revenue
FROM order_items oi
JOIN products p
    ON oi.product_id = p.product_id
GROUP BY p.product_category_name
ORDER BY total_revenue DESC
LIMIT 5;
```

EXPLANATION

- Joins order_items with products
- Sums total price per product category
- Sorts descending and limits results to the top 5

EXPECTED OUTPUT

product_category_name	total_revenue
health_beauty	R\$1.26M
watches_gifts	R\$1.21M

BUSINESS INSIGHT

Directs marketing budget and inventory investment toward the categories driving the most revenue.

Q9 Above-Average Sellers

9

Which sellers fulfill more orders than the average seller?

SQL QUERY

```
SELECT seller_id,
       COUNT(DISTINCT order_id) AS total_orders
FROM order_items
GROUP BY seller_id
HAVING COUNT(DISTINCT order_id) >
(
    SELECT AVG(order_count)
    FROM
    (
        SELECT COUNT(DISTINCT order_id) AS order_count
        FROM order_items
        GROUP BY seller_id
    ) AS avg_sellers
)
ORDER BY total_orders DESC;
```

EXPLANATION

- Counts fulfilled orders for every seller
- Calculates the average order count across all sellers
- Returns only sellers performing above that average

EXPECTED OUTPUT

seller_id	total_orders
abc123	542
xyz456	498

BUSINESS INSIGHT

Helps Amazon identify high-performing sellers for the Top Seller Badge program.

Q10 Review Scores by State

How does customer satisfaction vary by state for delivered orders?

SQL QUERY

```
SELECT c.customer_state,  
       ROUND(AVG(r.review_score),2) AS avg_review_score  
FROM orders o  
JOIN customers c  
     ON o.customer_id = c.customer_id  
JOIN order_reviews r  
     ON o.order_id = r.order_id  
WHERE o.order_status = 'delivered'  
GROUP BY c.customer_state  
ORDER BY avg_review_score DESC;
```

EXPLANATION

- Filters orders to only those marked 'delivered'
- Joins customers and order_reviews to each order
- Averages review scores grouped by customer state

EXPECTED OUTPUT

customer_state	avg_review_score
SP	4.2
RJ	3.9

BUSINESS INSIGHT

Pinpoints regions with lower post-delivery satisfaction, useful for logistics or service improvements.

Q11 Customers With No Reviews

Which customers placed orders but never left a review?

SQL QUERY

```
SELECT DISTINCT
    c.customer_unique_id,
    c.customer_id
FROM customers c
JOIN orders o
    ON c.customer_id = o.customer_id
LEFT JOIN order_reviews r
    ON o.order_id = r.order_id
WHERE r.review_id IS NULL;
```

EXPLANATION

- Left joins each customer's orders to order_reviews
- Filters to rows where no review exists
- Returns distinct customer identifiers

EXPECTED OUTPUT

customer_unique_id	customer_id
3b2a...	9f1c...
7e0d...	4a8b...

BUSINESS INSIGHT

Targets silent customers with review-request campaigns to improve feedback coverage and data quality.

Q12 Peak Order Month

1
2

Which single month generated the highest number of orders?

SQL QUERY

```
SELECT DATE_FORMAT(order_purchase_timestamp, '%Y-%m') AS
order_month,
        COUNT(*) AS total_orders
FROM orders
GROUP BY order_month
ORDER BY total_orders DESC
LIMIT 1;
```

EXPLANATION

- Formats purchase timestamps into year-month periods
- Counts total orders placed within each month
- Sorts descending and returns only the top month

EXPECTED OUTPUT

order_month	total_orders
2017-11	7,544

BUSINESS INSIGHT

Identifies the peak demand season (Black Friday) for staffing, inventory, and campaign planning.

Q13 Median Delivery Time

1
3

What is the median number of days between order placement and actual delivery?

SQL QUERY

```
WITH delivery_times AS (  
    SELECT  
        DATEDIFF(order_delivered_customer_date,  
                 order_purchase_timestamp) AS delivery_days  
    FROM orders  
    WHERE order_status = 'delivered'  
    AND order_delivered_customer_date IS NOT NULL  
)  
ranked AS (  
    SELECT  
        delivery_days,  
        ROW_NUMBER() OVER (ORDER BY delivery_days) AS rn,  
        COUNT(*) OVER () AS total_count  
    FROM delivery_times  
)  
SELECT  
    ROUND(AVG(delivery_days), 2) AS median_delivery_days  
FROM ranked  
WHERE rn IN (FLOOR((total_count + 1) / 2), CEIL((total_count + 1) / 2));
```

EXPLANATION

- Calculates delivery days per order with DATEDIFF
- Ranks delivery times and counts total delivered orders
- Averages the one or two middle values to get the median

EXPECTED OUTPUT

median_delivery_days
8.0

BUSINESS INSIGHT

Gives logistics leadership a true typical delivery speed, unskewed by rare remote-area delays that would distort a simple average.

Q14 Products Never Ordered

1
4

Which products have never been ordered by a customer?

SQL QUERY

```
SELECT
    p.product_id,
    p.product_category_name
FROM products p
LEFT JOIN order_items oi
    ON p.product_id = oi.product_id
WHERE oi.order_id IS NULL;
```

EXPLANATION

- Left joins products to order_items
- Filters to products with no matching order rows
- Returns the full list of never-ordered products

EXPECTED OUTPUT

product_id	product_category_name
e5f2...	toys
9c1a...	garden_tools

BUSINESS INSIGHT

Flags dead listings for the catalog team to review, reprice, or remove during quarterly clean-up.